

Predicting 30-Day Hospital Readmission

& AI Fairness Audit for Diabetic Patients

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Motivation & Problem

Which diabetic patients will return within 30 days — and does our model treat all racial groups fairly?

Why it matters:

- 30-day readmission rate is a standard metric for assessing the quality of patient care.
It indicates whether a patient was discharged too early or without adequate follow-up plans.
- A predictive model that identifies high-risk patients enables targeted post-discharge interventions.

The fairness question:

If the model systematically misses high-risk minority patients, it amplifies existing healthcare disparities.

Prior work: Strack et al. (2014) – HbA1c & readmission; Obermeyer et al. (2019) – racial bias in healthcare AI

Dataset Overview

UCI Diabetes 130-US Hospitals (1999–2008)

101,766

Patient Records

47

Features

11.16%

Readmitted <30 days
(Positive class)

130

US Hospitals

Features:

- Demographics: race, age, gender
- Admission details: source, discharge disposition
- Clinical measurements: medications, time in hospital
- 24 medication columns
- prior visit counts

Target:

- 3-class readmission: <30, >30, No
→ binary (readmitted <30 days vs. otherwise)

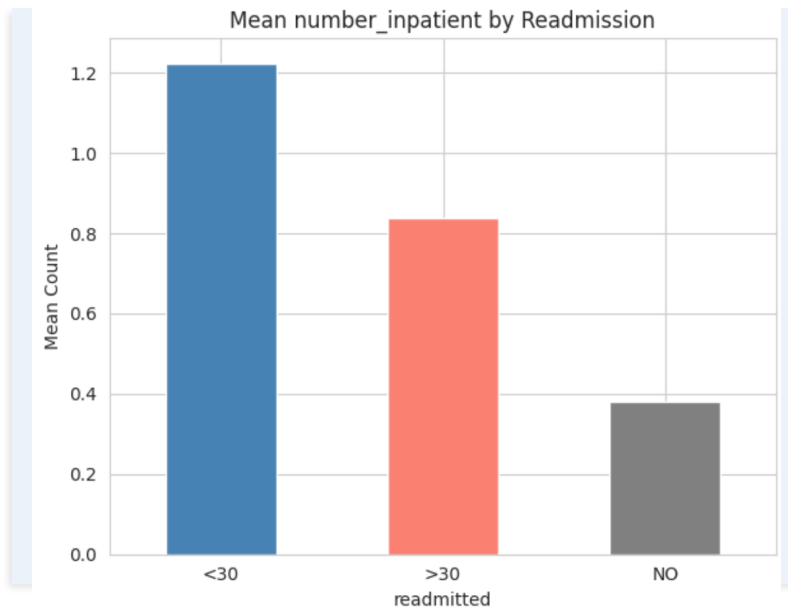
EDA Highlights

Key Findings

- Prior inpatient visits: highest in <30 readmissions
- Prior emergency visits: highest in <30 readmissions
- Time in hospital & medications: weak discriminators
- Race/gender: similar readmission proportions
- K-Means (k=3): distinct utilization profiles

Pre-processing

Dropped 4 high-missing columns • Imputed race with mode • One-hot encoded categorical features: 43 → 2,365



Modeling Approach

Random Forest Classifier (200 trees)

Class Imbalance

class_weight='balanced'
Upweights minority class
error

Regularization

max_depth=15
min_samples_leaf=20

Train/Test Split

Stratified 80/20
Preserves 11.16% positive
rate

Why RF?

Well-suited for 2,365 features, requires no scaling, offers feature importance interpretability, and is robust to overfitting through ensemble averaging.

Evaluation: Classification report, ROC curve + AUC, Precision-Recall curve + AP, confusion matrix, feature importances.

Model Results

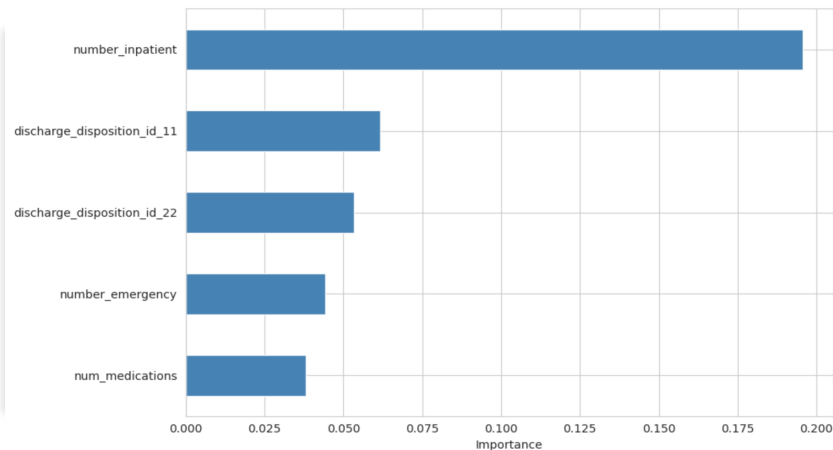
Metric	Value
ROC-AUC	0.672
Average Precision	0.216
Recall (<30)	58%
Precision (<30)	18%

Key Insight

~2x better than random (AP 0.22 vs 0.11 baseline).
PR curve reveals limitations ROC obscures.

Top Features

1. **number_inpatient (0.19)**
2. discharge_disposition_id_11 (0.06)
3. discharge_disposition_id_22 (0.05)
4. number_emergency (0.04)
5. num_medications (0.03)

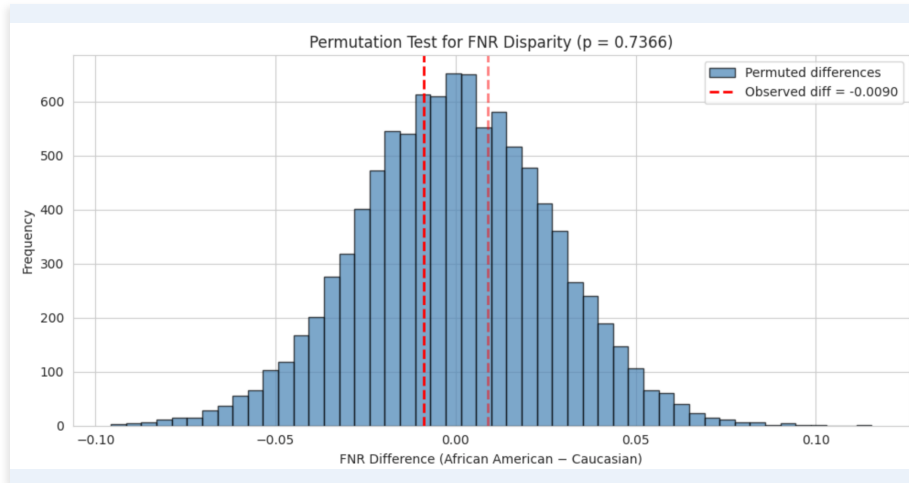


AI Fairness Audit

Does the model miss high-risk patients unequally?

False Negative Rate by Race

Race	FNR	N (pos)
African American	0.411	433
Caucasian	0.420	1,726
Hispanic	0.449	49
Asian	0.471	17
Other	0.440	25



Permutation Test: $p = 0.75$ (10,000 iterations)

No significant racial bias. FNR gap (-0.009) is indistinguishable from random noise (H_0 not rejected, $\alpha = 0.05$).

Conclusions & Future Work

Key Takeaways

- Prior hospitalization is the strongest predictor of 30-day readmission
- Model is ~2x better than random, but precision too low for direct clinical deployment
- No racial bias detected in model errors (permutation test $p = 0.75$)

Future Directions

- Employ the Synthetic Minority Over-sampling Technique (SMOTE)
- Integrate natural language processing to parse the actual doctors' free-text triage notes

Thank You — Questions?